ALGEBRA 1 · FACTORING AND ZERO PRODUCT PROPERTY

# Factor Rectangle Builder

Explore & Reflect **TEACHER KEY**

Name _____ Date _____ Period _____

Open **Factor Rectangle Builder** on Mr. K's Classroom: Algebra 1 → Factoring and Zero Product Property → Interactive Math. Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3, then do Part 4 on your own.

## 1 Predict before you touch the tool

1. To factor $x^2 + 7x + 12$, what two numbers are you looking for?

   **Two numbers that multiply to 12 and add to 7: 3 and 4.**

## 2 Explore use the tool and record what you see

**Factor each trinomial**

Find the rectangle, then read the zeros from the graph.

| Trinomial | Two numbers | Factored form | Zeros (x-intercepts) |
|---|---|---|---|
| $x^2 + 5x + 6$ | **2 and 3** | **$(x + 2)(x + 3)$** | **–2 and –3** |
| $x^2 + 7x + 12$ | **3 and 4** | **$(x + 3)(x + 4)$** | **–3 and –4** |
| $x^2 – x – 6$ | **–3 and 2** | **$(x – 3)(x + 2)$** | **3 and –2** |
| $x^2 – 8x + 15$ | **–3 and –5** | **$(x – 3)(x – 5)$** | **3 and 5** |
| $x^2 + 2x – 8$ | **4 and –2** | **$(x + 4)(x – 2)$** | **–4 and 2** |
| $x^2 + 6x + 9$ | **3 and 3** | **$(x + 3)^2$** | **–3 (touches)** |
| $x^2 – 9$ | **3 and –3** | **$(x + 3)(x – 3)$** | **–3 and 3** |
| $x^2 + x + 1$ | **none** | **not factorable** | **no real zeros** |

## 3 Write the rule fill in each blank from what you observed

1. To factor $x^2 + bx + c$, find two numbers that multiply to **c** and add to **b**.
2. If c is negative, the two numbers have **opposite** signs.
3. Zero product property: if $(x – p)(x – q) = 0$, then $x =$ **p** or $x =$ **q**.
4. The zeros of a trinomial are the **x-intercepts** of its graph.

## 4 Check no tool: show your work

1. Factor $x^2 + 9x + 20$.

   **$(x + 4)(x + 5)$**

2. Factor $x^2 – 2x – 15$.

   **$(x – 5)(x + 3)$**

3. Solve $x^2 + x – 12 = 0$.

   **$x = –4$ or $x = 3$**

4. Find the x-intercepts of $y = x^2 - 6x + 8$.

**(2, 0) and (4, 0)**

## 5 Reflect

Why are the zeros the opposite signs of the numbers in the factors?

**For (x + 3) = 0 we need x = –3: the zero is the value that makes the factor equal 0.**